TUTORIAL 7

Determinant Properties, Cofactor Expansion, and Cramer's Rule

MOTIVATION AND GUIDING IDEA

By the end of this tutorial, you will be able to choose an efficient determinant calculation, track the effect of row operations, identify conditions for invertibility, and connect algebraic structure with changes of basis and area.

NOTATION AND CONVENTIONS

The set $F[X]$ consists of polynomials in X with coefficients in F. The degree of a nonzero polynomial is the largest power with a nonzero coefficient. If $p(a) = 0$, division by $X - a$ has constant remainder $p(a) = 0$, so $p(X) = (X - a)q(X)$. Induction on the degree therefore shows that a nonzero polynomial of degree d has at most d distinct roots. These are the facts used in the Vandermonde and Cauchy calculations below. For a matrix A with n columns, we also write $\ker A = \{x \in F^n : Ax = 0\}$ for its homogeneous solution set.

EXERCISE 7.1

Please compute the determinants using row operations, stating the effect of each operation on the determinant.

1. For $t \in \mathbb{R}$,
$$A(t) = \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 2 & 1 & t & 9 \end{pmatrix}.$$
2.
$$B = \begin{pmatrix} 2 & 1 & 0 & 3 & -1 \\ 0 & 1 & 2 & -1 & 1 \\ 2 & -2 & -6 & 6 & -4 \\ 1 & 0 & 1 & 0 & 2 \\ 0 & 2 & 0 & 1 & 1 \end{pmatrix}.$$

WORKED SOLUTION

1. Add multiples of other rows to the fourth row; these operations preserve the determi-

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EXERCISE 7.1 CONTINUED

nant:

$$A(t) \xrightarrow{R_4 \leftarrow R_4 - 2R_1} \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & -1 & t & 11 \end{pmatrix}$$

$$\xrightarrow{R_4 \leftarrow R_4 + R_2} \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & t+2 & 12 \end{pmatrix}$$

$$\xrightarrow{R_4 \leftarrow R_4 - (t+2)R_3} \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 8-2t \end{pmatrix}.$$

Multiplying the diagonal entries gives $\det A(t) = 1 \cdot 1 \cdot 1 \cdot (8 - 2t) = 8 - 2t$.

2. The rows of B satisfy $r_3 = r_1 - 3r_2$. Thus the two operations $R_3 \leftarrow R_3 - R_1$ and $R_3 \leftarrow R_3 + 3R_2$ produce a zero row without changing the determinant. A matrix with a zero row has determinant zero, so $\det B = 0$.

RECALL AND TOOLS

Basic properties (from the lectures):

- **Multiplicativity:** For every $A, B \in M_n(\mathbb{R})$,

$$\det(AB) = \det(A)\det(B).$$

- **Failure of additivity (in general):** In general, the following equality does not hold:

$$\det(A + B) = \det(A) + \det(B).$$

- **Invariance under transposition:** For every $A \in M_n(\mathbb{R})$,

$$\det(A^T) = \det(A).$$

- **Scaling the entire matrix:** For every $c \in \mathbb{R}$ and $A \in M_n(\mathbb{R})$,

$$\det(cA) = c^n \det(A),$$

where n is the size of the matrix.

Fundamental theorem:

PROPOSITION 7.1

For every $A \in M_n(\mathbb{R})$, A is invertible if and only if $\det(A) \neq 0$.

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EXERCISE 7.2

Let $A, B \in M_n(\mathbb{R})$, with A singular and B invertible. Please prove or disprove that the matrix in each part is singular:

1. $A^T B$.
2. $B^{-1}AB$.
3. $A + A^2 + A^3 + A^4$.
4. $(A + I)^2 - (A - I)^2$.
5. $(A + B)^2 + (A - B)^2$.

WORKED SOLUTION

Use $\det A = 0$, $\det B \neq 0$, and multiplicativity of the determinant.

1. $\det(A^T B) = \det(A^T)\det B = \det A \det B = 0$.
2. $\det(B^{-1}AB) = \det(B^{-1})\det A \det B = 0$.
3. Factor out A on the left:

$$A + A^2 + A^3 + A^4 = A(I + A + A^2 + A^3),$$

Therefore the determinant is zero.

4. Since $AI = IA = A$, we obtain

$$(A + I)^2 - (A - I)^2 = (A^2 + 2A + I) - (A^2 - 2A + I) = 4A.$$

Consequently, $\det(4A) = 4^n \det A = 0$.

5. Expand while preserving the order of multiplication:

$$(A + B)^2 + (A - B)^2 = (A^2 + AB + BA + B^2) + (A^2 - AB - BA + B^2) = 2(A^2 + B^2).$$

The determinant is therefore $2^n \det(A^2 + B^2)$, but it need not vanish. A counterexample is $A = 0$, $B = 2I_n$: the resulting matrix is $8I_n$, which is invertible and has determinant $8^n \neq 0$.

7.1 Further Determinant Expansions and Properties

7.1.1 The Determinant Formula

RECALL AND TOOLS

Let $A = (a_{ij}) \in M_n(\mathbb{R})$ be a square matrix of size n. For $1 \leq i, j \leq n$, the **minor matrix** corresponding to entry (i, j), denoted M_{ij}, is the square matrix of size $n - 1$ obtained by deleting row i and column j from A.

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DEFINITION 7.1

For every $n \geq 1$, the determinant is the unique function $\det : M_n(\mathbb{F}) \to \mathbb{F}$ that is linear in each row separately with the other rows fixed, vanishes when two rows are equal, and satisfies $\det I_n = 1$. Row operation rules and cofactor expansions follow. Expansion along row i gives $\det A = \sum_{j=1}^{n} a_{ij} C_{ij}$; the result is independent of the selected row. With the convention $\det(\varnothing) = 1$ for the empty matrix of size 0, the formula also holds for $n = 1$.

NOTE

For square matrices over any field, $\det(AB) = \det A \det B$, $\det(A^T) = \det A$, and $\det(cA) = c^n \det A$. Also, A is invertible exactly when $\det A \neq 0$; then $\det(A^{-1}) = (\det A)^{-1}$. The determinant is not generally additive: $\det(A + B) \neq \det A + \det B$ in general.

EXAMPLE 7.1

Let $A = \begin{pmatrix} 4 & -1 & 3 \\ 2 & 5 & 0 \\ -2 & 1 & 6 \end{pmatrix}$. Then $M_{11} = \begin{pmatrix} 5 & 0 \\ 1 & 6 \end{pmatrix}$ and $M_{23} = \begin{pmatrix} 4 & -1 \\ -2 & 1 \end{pmatrix}$.

DEFINITION 7.2

Let $A = (a_{ij}) \in M_n(\mathbb{R})$ be square of size n. Its **determinant**, denoted $\det(A)$, can be computed recursively in n:

1. If $n = 1$, then $A = (a_{11})$ and we define
$$\det(A) = a_{11}.$$
2. If $n = 2$, then
$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \qquad \det(A) = a_{11}a_{22} - a_{12}a_{21}.$$
3. If $n \geq 3$, then for every $1 \leq i \leq n$, expansion along row i gives
$$\det(A) = \sum_{j=1}^{n} a_{ij} \, (-1)^{i+j} \, \det(M_{ij}),$$
where M_{ij} is obtained by deleting row i and column j.
Equivalently, for every $1 \leq j \leq n$, expansion along column j gives
$$\det(A) = \sum_{i=1}^{n} a_{ij} \, (-1)^{i+j} \, \det(M_{ij}).$$

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EXERCISE 7.3

Please compute the determinants again, this time using cofactor expansion and its consequences.

1. For $t \in \mathbb{R}$,
$$A(t) = \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 2 & 1 & t & 9 \end{pmatrix}.$$
2.
$$B = \begin{pmatrix} 2 & 1 & 0 & 3 & -1 \\ 0 & 1 & 2 & -1 & 1 \\ 2 & -2 & -6 & 6 & -4 \\ 1 & 0 & 1 & 0 & 2 \\ 0 & 2 & 0 & 1 & 1 \end{pmatrix}.$$

WORKED SOLUTION

1. Expand along the first column. Its two nonzero entries are in positions $(1,1)$ and $(4,1)$, so
$$\det A(t) = \det M_{11} - 2\det M_{41}.$$
The minor matrices are
$$M_{11} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 1 & t & 9 \end{pmatrix}, \qquad M_{41} = \begin{pmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$
Expand the first along its first column and the second along its first row:
$$\det M_{11} = (1 \cdot 9 - 2t) + (2 \cdot 2 - 1 \cdot 1) = 12 - 2t,$$
$$\det M_{41} = 1(2 \cdot 2 - 1 \cdot 1) + (-1)(1 \cdot 1 - 2 \cdot 0) = 3 - 1 = 2.$$
Thus $\det A(t) = 12 - 2t - 2 \cdot 2 = 8 - 2t$, as obtained by the previous method.
2. Regard the determinant as a function of the rows. Since $r_3 = r_1 - 3r_2$, linearity in the third row gives
$$\begin{aligned}\det(r_1, r_2, r_1 - 3r_2, r_4, r_5) = {} & \det(r_1, r_2, r_1, r_4, r_5) \\ & - 3\det(r_1, r_2, r_2, r_4, r_5) = 0.\end{aligned}$$
Each term has two equal rows. Linearity in a row and vanishing for equal rows follow from the determinant expansion, so again $\det B = 0$.

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EXERCISE 7.4

Please find all $k \in \mathbb{R}$ for which

$$A(k) = \begin{pmatrix} 2 & k & 4 \\ 3 & k+1 & 6 \\ -1 & -1 & k+3 \end{pmatrix}$$

is singular, using its determinant.

WORKED SOLUTION

Expanding along the first row gives

$$\begin{aligned} \det A(k) &= 2\big((k+1)(k+3)+6\big) - k\big(3(k+3)+6\big) + 4(k-2) \\ &= 2k^2 + 8k + 18 - 3k^2 - 15k + 4k - 8 \\ &= -k^2 - 3k + 10 = -(k+5)(k-2). \end{aligned}$$

Thus the matrix is singular exactly when $k = -5$ or $k = 2$.

EXERCISE 7.5

1. Let $n \geq 2$ and let $x_1, \ldots, x_n$ be real (or complex) numbers. Define the Vandermonde matrix

$$V(x_1, \ldots, x_n) = \left(x_i^{j-1}\right)_{i,j=1}^{n} = \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{pmatrix}.$$

Please prove that

$$\det\big(V(x_1, \ldots, x_n)\big) = \prod_{1 \leq i < j \leq n} (x_j - x_i).$$

WORKED SOLUTION

1. Write

$$\Delta_n(x_1, \ldots, x_n) = \det\big(V(x_1, \ldots, x_n)\big).$$

We prove by induction on n that

$$\Delta_n(x_1, \ldots, x_n) = \prod_{1 \leq i < j \leq n} (x_j - x_i).$$

Base case, $n = 2$:

$$\Delta_2(x_1, x_2) = \det \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \end{pmatrix} = x_2 - x_1,$$

as required.

Inductive step: assume the statement holds for $n-1$, and prove it for n. Regard Δ_n as a polynomial in x_n, with $x_1, \ldots, x_{n-1}$ fixed. Expanding the determinant along row n gives

$$\Delta_n(x_1, \ldots, x_n) = \sum_{j=1}^{n} x_n^{j-1} \, (-1)^{n+j} \, \det(M_{nj}),$$

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EXERCISE 7.5 CONTINUED

Thus Δ_n is a polynomial in x_n of degree at most $n-1$. The only term containing x_n^{n-1} comes from column n, so its coefficient is

$$(-1)^{n+n}\det(M_{nn}) = \det\big(V(x_1,\dots,x_{n-1})\big) = \Delta_{n-1}(x_1,\dots,x_{n-1}).$$

If two of $x_1,\dots,x_{n-1}$ are equal, the matrix has two equal rows, and both sides of the desired formula are zero. Hence we may assume these numbers are pairwise distinct. For every $k \in \{1,\dots,n-1\}$, substituting $x_n = x_k$ makes rows k and n of V equal, so

$$\Delta_n(x_1,\dots,x_{n-1},x_k) = 0.$$

Therefore $(x_n - x_k)$ divides Δ_n for every $k = 1,\dots,n-1$ (as a polynomial in x_n). Since these roots are distinct and the degree is at most $n-1$, there is a polynomial P such that

$$\Delta_n(x_1,\dots,x_n) = \Big(\prod_{k=1}^{n-1}(x_n - x_k)\Big)\,P(x_1,\dots,x_{n-1}),$$

where P is independent of x_n.

Compare the coefficients of x_n^{n-1} on both sides. On the left the coefficient is $\Delta_{n-1}(x_1,\dots,x_{n-1})$; on the right the coefficient of x_n^{n-1} in $\prod_{k=1}^{n-1}(x_n - x_k)$ is 1. Hence

$$P(x_1,\dots,x_{n-1}) = \Delta_{n-1}(x_1,\dots,x_{n-1}).$$

It follows that

$$\Delta_n(x_1,\dots,x_n) = \Delta_{n-1}(x_1,\dots,x_{n-1})\prod_{k=1}^{n-1}(x_n - x_k).$$

Finally, by the induction hypothesis,

$$\Delta_{n-1}(x_1,\dots,x_{n-1}) = \prod_{1\le i<j\le n-1}(x_j - x_i),$$

and therefore

$$\Delta_n(x_1,\dots,x_n) = \Big(\prod_{1\le i<j\le n-1}(x_j - x_i)\Big)\Big(\prod_{k=1}^{n-1}(x_n - x_k)\Big) = \prod_{1\le i<j\le n}(x_j - x_i).$$

This proves the formula.

Additional practice

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EXERCISE 7.6 CHALLENGE

Let $A, B \in M_n(\mathbb{R})$ be invertible and satisfy $AB = -BA$. Please prove that n is even. For every $m \geq 1$, please construct such a pair of size $2m$ with $A^2 = B^2 = I_{2m}$.

WORKED SOLUTION

Determinant rules give

$$\det A \det B = \det(AB) = \det(-BA) = (-1)^n \det B \det A.$$

Both determinants are nonzero, so cancellation gives $1 = (-1)^n$. Over $\mathbb{R}$, n must be even. For every $m \geq 1$, take

$$A = \begin{pmatrix} I_m & 0 \\ 0 & -I_m \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & I_m \\ I_m & 0 \end{pmatrix}.$$

Direct block multiplication gives

$$AB = \begin{pmatrix} 0 & I_m \\ -I_m & 0 \end{pmatrix}, \qquad BA = \begin{pmatrix} 0 & -I_m \\ I_m & 0 \end{pmatrix} = -AB,$$

and $A^2 = B^2 = I_{2m}$. Thus both are invertible.

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EXERCISE 7.7 CHALLENGE

Let $n \geq 1$, and let $x_1, \ldots, x_n$ be distinct elements of a field F, and let $y_1, \ldots, y_n$ be distinct. Assume $x_i + y_j \neq 0$ for every i, j, and define

$$C = \left(\frac{1}{x_i + y_j} \right)_{i,j=1}^{n}.$$

Please prove that every square submatrix of C is invertible. Please deduce that the real matrix $H_n = (1/(i+j-1))_{i,j=1}^{n}$ is invertible for every $n \geq 1$.

WORKED SOLUTION

First suppose $Cc = 0$, and define

$$q(t) = \sum_{j=1}^{n} c_j \prod_{k \neq j} (t + y_k).$$

This polynomial has degree at most $n-1$. Row i of the equation gives

$$q(x_i) = \left(\prod_{k=1}^{n} (x_i + y_k) \right) \sum_{j=1}^{n} \frac{c_j}{x_i + y_j} = 0.$$

It has n distinct roots and must therefore be the zero polynomial. At $t = -y_j$, every term except term j vanishes, so

$$0 = q(-y_j) = c_j \prod_{k \neq j} (y_k - y_j).$$

The product is nonzero, forcing $c_j = 0$ for every j. Thus C is invertible. A square submatrix selects equally many row and column indices. The selected elements remain distinct within each list, and their sums are still nonzero, so the same proof applies. For the final assertion, take $x_i = i - 1$ and $y_j = j$ over $\mathbb{R}$. Every denominator $i + j - 1$ is positive, and $C = H_n$.

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EXERCISE 7.8 EXAM QUESTION

Please work in $\mathbb{Z}_7$. Please solve the system in terms of $\omega \in \mathbb{Z}_7$:

$$\begin{cases} x + y + z = 2, \\ x + \omega y + z = 5, \\ x + y + (\omega + 1) z = 2. \end{cases}$$

Please write the solution set and number of solutions in every case.

WORKED SOLUTION

Subtract the first equation from each of the other two:

$$\begin{cases} x + y + z = 2, \\ (\omega - 1) y = 3, \\ \omega z = 0. \end{cases}$$

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EXERCISE 7.8 EXAM QUESTION CONTINUED

1. If $\omega = 1$, the equation $0 = 3$ gives no solution.
2. If $\omega = 0$, then $-y = 3$, so $y = 4$. Choosing $z = t$ gives
$$S_0 = \{(5 - t, 4, t) : t \in \mathbb{Z}_7\}.$$
There are exactly seven solutions, since different values of t give different third coordinates.
3. If $\omega \in \{2, 3, 4, 5, 6\}$, both ω and $\omega - 1$ are invertible. Thus
$$z = 0, \qquad y = 3(\omega - 1)^{-1}, \qquad x = 2 - y.$$
Each case has a unique solution:

ω	2	3	4	5	6
x	6	4	1	3	0
y	3	5	1	6	2
z	0	0	0	0	0

As a check, the coefficient determinant is $\omega(\omega - 1)$, which vanishes only at $\omega = 0, 1$.

7.2 Cramer's Rule

NOTATION AND CONVENTIONS

Write $A = (a_1 \ \cdots \ a_n)$, where a_j is column j of A. For $b \in F^n$, let $A_i(b)$ be the matrix obtained by replacing column i of A with b.

PROPOSITION 7.2

Let $A \in M_n(F)$ with $\det A \neq 0$. The unique solution of $Ax = b$ is
$$x_i = \frac{\det A_i(b)}{\det A}, \qquad i = 1, \ldots, n.$$

Proof. Invertibility gives a unique solution. Matrix-vector multiplication gives
$$b = Ax = \sum_{j=1}^{n} x_j a_j.$$
By linearity of the determinant in column i,
$$\det A_i(b) = \sum_{j=1}^{n} x_j \det A_i(a_j).$$
When $j \neq i$, two columns are equal, so that term vanishes. When $j = i$, the matrix is A. Thus $\det A_i(b) = x_i \det A$. Dividing by $\det A \neq 0$ proves the formula.

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EXAMPLE 7.2

We solve $Ax = b$ over $\mathbb{R}$, where

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 0 \\ 2 \\ 7 \end{pmatrix}.$$

Expansion in the first row gives $\det A = 1 - 2(0 - 2) = 5$. Replacing the columns gives

$$\det A_1(b) = \det \begin{pmatrix} 0 & 2 & 0 \\ 2 & 1 & 1 \\ 7 & 0 & 1 \end{pmatrix} = -2(2 - 7) = 10,$$

$$\det A_2(b) = \det \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 2 & 7 & 1 \end{pmatrix} = 2 - 7 = -5,$$

$$\det A_3(b) = \det \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \\ 2 & 0 & 7 \end{pmatrix} = 7 - 2(0 - 4) = 15.$$

Hence $x = (10/5, -5/5, 15/5)^T = (2, -1, 3)^T$. Direct substitution gives

$$Ax = \begin{pmatrix} 2 - 2 \\ -1 + 3 \\ 4 + 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ 7 \end{pmatrix} = b.$$

EXERCISE 7.9

Let $a_0, \ldots, a_n \in F^n$. Let D_j be the determinant obtained from columns $a_0, \ldots, a_n$ by omitting a_j and keeping the order of the other columns. Please prove that

$$\sum_{j=0}^{n} (-1)^j D_j a_j = 0.$$

WORKED SOLUTION

Let a_{ij} be entry i of a_j. For each $i = 1, \ldots, n$, form

$$C_i = \begin{pmatrix} a_{i0} & a_{i1} & \cdots & a_{in} \\ a_{10} & a_{11} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{n0} & a_{n1} & \cdots & a_{nn} \end{pmatrix},$$

where the bottom n rows are the matrix $(a_0 \ \cdots \ a_n)$. The top row equals row i of the lower block, so $\det C_i = 0$. Expanding in the top row gives

$$0 = \sum_{j=0}^{n} (-1)^{1+(j+1)} a_{ij} D_j = \sum_{j=0}^{n} (-1)^j D_j a_{ij}.$$

This is entry i of the required sum. Every entry is zero, so the sum is zero.

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EXERCISE 7.9 CONTINUED

For example, over $\mathbb{R}$, take

$$a_0 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad a_1 = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \quad a_2 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

Then $D_0 = 11, D_1 = -2, D_2 = -5$, and

$$11a_0 + 2a_1 - 5a_2 = \begin{pmatrix} 11 + 4 - 15 \\ 22 - 2 - 20 \end{pmatrix} = 0.$$

If at least one minor is nonzero, the identity gives an explicit nontrivial dependence among the columns.

EXERCISE 7.10

Let A be an invertible real $n \times n$ matrix with integer entries. Please prove that the following conditions are equivalent:

1. For every $b \in \mathbb{Z}^n$, the solution of $Ax = b$ belongs to $\mathbb{Z}^n$.
2. $\det A \in \{1, -1\}$.

Here $\mathbb{Z}^n$ is the set of columns with integer entries.

WORKED SOLUTION

If $\det A = 1$ or -1, then for every $b \in \mathbb{Z}^n$ all entries of $A_i(b)$ are integers. Their determinants are integers, and Cramer's rule gives

$$x_i = \frac{\det A_i(b)}{\det A} \in \mathbb{Z}.$$

Conversely, choose $b = e_j$ in turn, where e_j has entry 1 in position j and zeros elsewhere. The solution is column j of A^{-1}, so every entry of A^{-1} is an integer. Thus both $\det A$ and $\det A^{-1}$ are integers. Multiplicativity gives

$$(\det A)(\det A^{-1}) = \det I_n = 1.$$

Two integers with product 1 are both 1 or both -1, so $\det A \in \{1, -1\}$.

For example, $\begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$ has determinant 1, so every integer right-hand side gives an integer solution. In contrast, for $A = \operatorname{diag}(2, 1)$ and $b = (1, 0)^T$, the solution is $(1/2, 0)^T$.

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